LEE HYESUNG

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RESEARCH OBJECTIVE

The goal of my research is to enable collaborative and adaptive behaviors in multi-agent robotic systems by integrating learning-based control with analytical modeling.

EDUCATION

Eidgenössische Technische Hochschule Zürich (ETH Zürich) — Sep. 2026 — (Present)
Robotics, Systems, and Control M.Sc. — *Zürich, Switzerland*

Seoul National University (SNU) — Mar. 2020 — Aug. 2026
BS in Mechanical Engineering, Computer Science and Engineering — ** two years absence for military service*
GPA: 4.07/4.30 (3.92/4.00), — *Seoul, South Korea*
Graduate with Summa Cum Laude

Eidgenössische Technische Hochschule Zürich (ETH Zürich) — Feb. 2024 — Aug. 2024
Exchange Student in Mechanical Engineering Department — *Zürich, Switzerland*

Gwangju Science Academy for gifted (GSA) — Feb. 2017 — Jan. 2019
Top 5 students award among graduates — *Gwangju, South Korea*
GPA: 4.19/4.30

RESEARCH EXPERIENCES

Interactive & Networked Robotics Laboratory | SNU — Mar. 2025 — Dec. 2025
Undergraduate Researcher, Supervisors: Prof. Yongseok Lee and Prof. Dongjun Lee — *Seoul, South Korea*

- Designed sim-to-real object reconstruction methods with joint articulation
- Run articulated assets in simulation for manipulation tasks.

AI Research Team | Neuromeka Co., Ltd. — Jan. 2025 — Apr. 2025
Research Intern, Supervisor: Dr. Joonho Lee — *Seoul, South Korea*

- Design safe master-slave teleoperation system with haptic feedback and compliance control
- Trained collaborative robot with ACT, SERL policies for industrial application.

Computational Robotics Laboratory | ETH Zürich — Mar. 2024 — Aug. 2024
Semester Project, Supervisor: Prof. Dr. Stelian Coros — *Zürich, Switzerland*

- Train wheeled-quadrupedal robot, Unitree B2W, with PPO policy.
- Obtained various styles of stable locomotion from hierarchal control designs.

Bio Robotics Laboratory | SNU — Oct. 2023 — Jan. 2024
Undergraduate Research Intern, Supervisor: Prof. Kyujin Cho — *Seoul, South Korea*

- Designed and Manufactured penumatic elogation robot with controlled buckling.
- Applied origami structure on soft robot and modeled 3D folding kinematics mathematically.

COURSE PROJECTS

Mechatronics Competition | SNU — Spring 2025
Built a reading stand that scans, interpret, and turns the pages without destruction. — *Mechatronics*
3rd Place Award

Bridge design contest for the 3-point bending test | SNU — Spring 2025
Achieved a load resistance of 140 kg with 90 g carbon-fiber structure. — *Materials and Manufacturing Process*
1st Place Award

Jumping & Locomotive Robot Design | SNU — Fall 2020
Designed a rubber band–actuated jumping robot using a wooden-link structure. — *Creative Engineering Design*
Designed a wooden-link locomotive robot based on joint–link locomotion principles.

PUBLICATIONS

- **HS. Lee**, YS. Lee, KT. Lee, DJ. Lee, YS. Lee. RORA: Realistic Object Reconstruction with Articulation. arXiv:2608.04842, 2026. *(under review)* [Paper] [Project Page]

HONORS

National Presidential Science Scholarship | Republic of Korea 2020 — 2025
**Full coverage of 4 years tuition and stipend*

Talent Scholarship of Jeollanam-do Provincxse | Jeolla-namdo State 2020 — 2025
**Full coverage of 4 years talent development grant*

Han-sung Nobel Scholarship | Hanseong Sonjaehan Scholarship Foundation 2018

MEMBERSHIPS & ACTIVITIES

SNU Tomorrow's Engineering Membership (STEM) | SNU Mar. 2025 — Aug. 2026
Honor Society in College of Engineering, SNU
Operate international student event in Seoul
Held academic talk on robot learning frameworks

Bulnabi, Autonomous Flight Drone Club | College of Engineering, SNU Sep. 2024 — Dec. 2024
Acquired practical experience in drone hardware, control, and simulation-based validation

Military Open Source Hackathon | Ministry of Defense, Republic of Korea Sep. 2022 — Nov. 2022
Developed web application on real-time reporting & monitoring service of broken facilities

Military Service | Republic of Korea Army Mar. 2022 — Oct. 2023
**Army Sergeant, Honorable Discharge*

SNU Buddy, Official Student Organization of International Affairs | SNU Aug. 2021 — Feb. 2022
Organized networking events for incoming students, helped troublesome of foreign students.

TEACHING EXPERIENCES

Teaching Tutor
Calculus 1 | College of Engineering, SNU Winter 2023

Volunteering
Mathematic Class for underprivileged youth Winter 2023
Science Class for students preparing high school equivalency exam Winter 2021
Support of settlement and cultural adaptation for international students (Similar to ESN) Fall 2021
C Programming for middle school students Winter 2020

QUALIFICATIONS

English	TOEFL 103 *(My best score 108)*
German	German A1
Certificates	Information Processing Technician
Military	Army Sergeant, Honorable Discharge

SKILLS

Programming	C/C++, Python, Java, MATLAB, Simulink
Simulation	ROS1/2, Isaac Sim, Mujoco, Pytorch, Unity, Unreal
Graphics	Gaussian-Splatting
CAD	Fusion360, SolidWorks
Web Stack	TS-React, JavaScript, GCP